Universal LRH is impossible with an $n = 2$ response function

1 Setup

Activation space $\mathbb{R}^D$. Response function $R(v) = F(S(v))$ with F convex and monotonic, and

$$S(v) \;=\; \sum_{i=1}^{M} w_i (d_i^\top v)^2 \;=\; v^\top A v, \qquad A \;:=\; \sum_{i=1}^{M} w_i \, d_i d_i^\top \;\in\; \mathbb{R}^{D\times D}.$$

Here $\{d_i\}_{i=1}^M \subset \mathbb{R}^D$ are unit feature directions with $M > D$ (superposition; we assume the features span $\mathbb{R}^D$), and $w_i \geq 0$ are weights. The plateau is $\{v : S(v) < c\}$. Along a unit direction d, the perturbation magnitude that breaks the plateau is

$$m_d \;=\; \sqrt{c/q_d}, \qquad q_d \;:=\; d^\top A d.$$

Smaller $q_d \Leftrightarrow$ larger $m_d \Leftrightarrow$ plateau harder to break along d. Write $q_i := q_{d_i}$.

2 Universal LRH

Universal linear representation hypothesis: every feature direction is strictly more sensitive (strictly smaller m, strictly larger q) than every non-feature unit direction:

$$q_i \;>\; q_d \qquad \text{for every feature } i \text{ and every unit } d \notin \{d_1, \ldots, d_M\}.$$

3 Impossibility

Claim. For any weights $\{w_i\}_{i=1}^M$ and any threshold $c > 0$, universal LRH fails.

Proof. $A \succeq 0$. For any symmetric $A \in \mathbb{R}^{D\times D}$ with largest eigenvalue $\lambda_{\max}(A)$, the Rayleigh quotient $d \mapsto d^\top A d$ on the unit sphere attains its maximum exactly on the top eigenspace $V := \ker(\lambda_{\max}(A)\, I - A)$:

$$q_d \;=\; d^\top A d \;\leq\; \lambda_{\max}(A) \qquad \text{for every unit } d \in \mathbb{R}^D,$$

with equality iff $d \in V$. Let $r := \dim V \geq 1$.

Case $r < D$. The features span $\mathbb{R}^D$, so some feature $d_{i^*} \notin V$; hence $q_{i^*} < \lambda_{\max}(A)$. The unit sphere of V is infinite while the feature set is finite, so some unit $u \in V$ is not a feature direction. For that u, $q_u = \lambda_{\max}(A) > q_{i^*}$, contradicting universal LRH on feature i^*.

Case $r = D$. Then $A = \lambda_{\max}(A)\, I$ and $q_d = \lambda_{\max}(A)$ for every unit d, so the strict inequality fails for every feature.

Either way, universal LRH fails. □